

Table 52. I/O AC characteristics⁽¹⁾⁽²⁾ (continued)

Speed	Symbol	Parameter	Conditions	Min	Max	Unit
01	Fmax	Maximum frequency	C=50 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	10.00	MHz
			C=50 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	2.00	
			C=10 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	15.00	
			C=10 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	2.50	
	Tr/Tf	Output rise and fall time ⁽³⁾	C=50 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	30.00	ns
			C=50 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	60.00	
			C=10 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	15.00	
			C=10 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	30.00	
10	Fmax	Maximum frequency	C=50 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	30.00	MHz
			C=50 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	15.00	
			C=10 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	60.00 ⁽⁴⁾	
			C=10 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	30.00	
	Tr/Tf	Output rise and fall time ⁽³⁾	C=50 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	11.00	ns
			C=50 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	22.00	
			C=10 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	4.00	
			C=10 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	8.00	
11	Fmax	Maximum frequency	C=30 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	60.00 ⁽⁴⁾	MHz
			C=30 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	30.00	
			C=10 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	80.00 ⁽⁴⁾	
			C=10 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	40.00	
	Tr/Tf	Output rise and fall time ⁽³⁾	C=30 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	5.50	ns
			C=30 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	11.00	
			C=10 pF, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	-	2.50	
			C=10 pF, $2\text{ V} \leq V_{DD} \leq 2.7\text{ V}$	-	5.00	

1. The I/O speed is configured using the OSPEEDRy[1:0] bits. The Fm+ mode is configured in the SYSCFG_CFGR1 register. Refer to the RM0490 reference manual for a description of GPIO Port configuration register.

2. Specified by design – Not tested in production.

3. The fall time is defined between 70% and 30% of the output waveform, according to I²C specification.

4. This value represents the I/O capability but the maximum system frequency is limited to 48 MHz.